Mathematics practice booklet: Grade 10 expectations

Grade 10 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 10 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: A-APR.A.1, A-SSE.B.3, A-REI.B.4, F-IF.C.7, F-IF.C.8

A worked example

Solve by factoring: x^2 - 2x - 15 = 0. Then check one answer.

1. Look for two numbers that multiply to -15 and add to -2: they are -5 and 3.
2. Factor: (x - 5)(x + 3) = 0.
3. A product is zero only when one factor is zero, so x - 5 = 0 or x + 3 = 0.
4. Check x = 5: 25 - 10 - 15 = 0. It works.

x = 5 or x = -3

Questions

1. Which pair of values solves x^2 - 5x + 6 = 0?

   A. x = -2 or x = -3

   B. x = 1 or x = 6

   C. x = 2 or x = 3

   D. x = -1 or x = -6

2. A rectangular banner is x m wide and (x + 3) m long, and its area is 10 square m. Solve x^2 + 3x - 10 = 0 by factoring, then give the width and length of the banner.

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3. Which shows all the solutions of 2x^2 - 8 = 0?

   A. x = 2 only

   B. x = 2 or x = -2

   C. x = 4 or x = -4

   D. x = 4 only

4. Solve by completing the square: x^2 + 6x + 2 = 0

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5. Use the quadratic formula to solve x^2 - 4x - 1 = 0.

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6. Use the quadratic formula to solve $2x^2 + 3x - 2 = 0$. Which pair is correct?

   A. $x = 1/2$ or $x = 2$

   B. $x = -1/2$ or $x = 2$

   C. $x = -1/2$ or $x = -2$

   D. $x = 1/2$ or $x = -2$

7. Find the discriminant of $x^2 + 2x + 5 = 0$. How many real solutions does it have?

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8. Expand and simplify: $(x + 3)(2x - 5)$

   A. $2x^2 + x - 15$

   B. $2x^2 - 15$

   C. $2x^2 + 11x - 15$

   D. $2x^2 - x - 15$

9. Let $A = 3x^2 + 2x - 1$ and $B = x^2 - 5x + 4$. Find $A + B$ and $A - B$.

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10. For $y = x^2 - 6x + 5$, find the x-intercepts, the vertex and the y-intercept.

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11. The rectangular floor of a majlis is 3 m longer than it is wide, and its area is 40 square m. What are its width and length?

    A. Width 8 m, length 11 m

    B. Width 5 m, length 8 m

    C. Width 4 m, length 10 m

    D. Width 8 m, length 5 m

12. Omar kicks a ball upward. Its height is $h = -5t^2 + 20t$ metres after t seconds. When does it land, and what is its greatest height?

Section 2: Shapes, measuring and space

Standards practised: G-CO.B.8, G-CO.C.9, G-CO.C.10, G-SRT.B.5, G-SRT.C.6, G-SRT.C.8, G-C.A.2, G-C.B.5

A worked example

A ramp is 10 m long and rises at an angle of 20 degrees to the ground. How high does it rise? (sin 20 degrees is about 0.342)

1. The ramp is the hypotenuse. The height is the side opposite the 20 degree angle.
2. Opposite and hypotenuse go with sine: sin 20 = height / 10.
3. Height = 10 × 0.342 = 3.42.

About 3.4 m

Questions

1. Which information is NOT enough to prove two triangles congruent?

   A. Three pairs of equal sides (SSS)

   B. Two sides and the angle between them (SAS)

   C. Two angles and the side between them (ASA)

   D. Two sides and an angle not between them (SSA)

2. In triangles ABC and DEF, AB = DE = 6 cm, angle B = angle E = 50 degrees and BC = EF = 9 cm. Which criterion shows they are congruent?

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3. Two straight lines cross. One of the four angles is 125 degrees. What are the other three angles?

   A. 55, 55, 55

   B. 125, 125, 55

   C. 125, 55, 55

   D. 65, 65, 125

4. A triangular shade sail is isosceles and its top (apex) angle is 40 degrees. Find the two base angles and explain why they are equal.

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5. Triangle ABC has sides 3, 4 and 5 cm. Triangle DEF is similar with sides 9, 12 and x cm. What is x?

   A. 11

   B. 15

   C. 18

   D. 45

6. A lamp post 3 m tall casts a shadow 2 m long. At the same time a palm tree casts a shadow 10 m long. How tall is the palm tree?

   A. 15 m

   B. 11 m

   C. 20 m

   D. 30 m

7. A ramp is 10 m long and rises 6 m. Find the sine of the angle it makes with the ground, and the size of that angle to the nearest degree.

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8. Khalid stands 50 m from the base of a tower and sees the top at an angle of elevation of 35 degrees. Find the height of the tower. (tan 35 degrees is about 0.700)

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9. A rectangular courtyard is 12 m long and 5 m wide. How long is a walk along its diagonal?

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10. A circular garden has radius 7 m. Find its circumference and area in terms of pi, then to one decimal place. A quarter of the garden is a sector: find its arc length and its area to one decimal place.

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11. In a circle, a central angle of 80 degrees and an inscribed angle both stand on the same arc. Find the inscribed angle. What is an inscribed angle that stands on a diameter?

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12. A point P is 13 cm from the centre of a circle of radius 5 cm. A tangent from P touches the circle at T. How long is PT?

    A. 8 cm

    B. 18 cm

    C. 12 cm

    D. About 14 cm

Section 3: Data, graphs and chance

Standards practised: S-CP.A.1, S-CP.A.2, S-CP.A.3, S-CP.A.4, S-CP.B.6, S-CP.B.9

A worked example

A jar holds 3 red and 2 green sweets. Hind takes two sweets without putting the first one back. Find the probability that both are red.

1. The first sweet is red with probability 3/5.
2. Now 2 red sweets remain among 4 sweets, so the second is red with probability 2/4.
3. Multiply: 3/5 × 2/4 = 6/20 = 3/10.

3/10

Questions

1. A coin is tossed and a die is rolled. What is the probability of heads and a 4?

   A. 1/12

   B. 1/6

   C. 1/8

   D. 2/3

2. P(A) = 0.4, P(B) = 0.5 and P(A and B) = 0.2. Are A and B independent? Explain.

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3. A bag has 5 red and 3 blue counters. Two counters are drawn, and the first is put back before the second. Find the probability that both are red.

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4. Use the same bag, but this time the first counter is NOT put back. What is the probability that both are red?

   A. 25/64

   B. 5/16

   C. 5/14

   D. 4/7

5. 50 students were asked whether they prefer swimming or running. Boys: 15 swimming, 10 running. Girls: 20 swimming, 5 running. How many prefer swimming, and what is P(swimming) for a student chosen at random?

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6. In the same survey, a girl is chosen at random. What is the probability that she prefers swimming?

   A. 2/5

   B. 7/10

   C. 4/7

   D. 4/5

7. In the same survey, a student who prefers swimming is chosen at random. Find the probability that it is a girl.

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8. Using the same survey, are being a girl and preferring swimming independent? Explain.

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9. P(A) = 0.5 and P(A and B) = 0.3. What is P(B given A)?

   A. 0.15

   B. 0.6

   C. 0.3

   D. 0.8

10. Six runners are in a school race. In how many different ways can gold, silver and bronze be given?

    A. 18

    B. 20

    C. 216

    D. 120

11. How many different ways can the 4 letters of OMAN be arranged in a row? If they are arranged at random, what is the probability of spelling OMAN?

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12. The chance that a metro train is on time on any given day is 0.9, and days are independent. Find the probability that it is on time on both Sunday and Monday, and the probability that it is late at least once.

Section 4: Numbers and calculating

Standards practised: N-RN.A.1, N-RN.A.2, N-RN.B.3, N-Q.A.1, N-Q.A.3

A worked example

Simplify the square root of 48 plus the square root of 12.

1. Write each as a perfect square times another number: √48 = √16 x √3 = 4√3.
2. √12 = √4 x √3 = 2√3.
3. These are like terms, so add them: 4√3 + 2√3 = 6√3.

6√3

Questions

1. Which is √72 in simplest form?

   A. 6√2

   B. 36√2

   C. 9√2

   D. 8√3

2. Two square tiles have areas 50 square cm and 18 square cm. Find the sum of their side lengths in simplest radical form.

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3. Simplify: √3 x √12

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4. Write 6 / √3 with a whole-number denominator.

   A. 2

   B. 2√3

   C. 6√3

   D. 3√2

5. Evaluate: 49^(1/2) and 8^(1/3)

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6. Evaluate: 4^(-3/2)

   A. -8

   B. -6

   C. 1/8

   D. 8

7. Write the cube root of 5 using a rational exponent, and write 7^(2/5) using a radical.

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8. Which of these numbers is irrational?

   A. √2 x √8

   B. √49

   C. 0.25

   D. 1/2 + √3

9. A car travels at 108 km per hour. What is this speed in metres per second?

   A. 30 m/s

   B. 1.8 m/s

   C. 108 m/s

   D. 388.8 m/s

10. A desalination plant makes 3 × 10^4 cubic m of water a day, and 1 cubic m is 10^3 litres. How many litres is that in 250 days? Give the answer in scientific notation.

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11. A tank really holds 250 litres. Hamdan measures it as 245 litres. Find the percent error.

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12. A solar park is 2 km long and 500 m wide. Find its area in square metres and in square kilometres. Why should the units match before you multiply?

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. C. x = 2 or x = 3. Factor to (x - 2)(x - 3) = 0, so x = 2 or x = 3. If your child chose A (-2 or -3), they flipped the signs of the answers.
2. Factor: (x + 5)(x - 2) = 0, so x = 2 or x = -5. A width cannot be negative, so the banner is 2 m wide and 2 + 3 = 5 m long.
3. B. x = 2 or x = -2. 2x^2 = 8, so x^2 = 4, and x = 2 or x = -2. If your child chose A (2 only), they forgot the negative answer; if C (4 or -4), they stopped at x^2 = 4 and did not take the square root.
4. Move 2 to the other side: x^2 + 6x = -2. Add 9 to both sides to get (x + 3)^2 = 7, so x = -3 + √7 or x = -3 - √7.
5. Use x = (4 ± √20) / 2, because the discriminant is 16 + 4 = 20 and √20 = 2√5. This gives x = 2 + √5 or x = 2 - √5.
6. D. x = 1/2 or x = -2. The discriminant is 9 + 16 = 25, so x = (-3 ± 5) / 4, which gives 1/2 or -2. If your child chose A (1/2 or 2), they lost the minus sign in (-3 - 5) / 4 = -2.
7. Discriminant = 2^2 - 4 × 1 × 5 = 4 - 20 = -16. It is negative, so there are no real solutions.
8. A. 2x^2 + x - 15. Multiply every pair: 2x^2 - 5x + 6x - 15 = 2x^2 + x - 15. If your child chose B (2x^2 - 15), they multiplied only the first terms and the last terms.
9. Add like terms: 3x^2 + x^2 = 4x^2, 2x - 5x = -3x and -1 + 4 = 3, so A + B = 4x^2 - 3x + 3. Subtract every term of B: 2x^2 + 7x - 5.
10. Factor: (x - 1)(x - 5) = 0 gives x-intercepts 1 and 5. The vertex is at x = 3, and y = 9 - 18 + 5 = -4, so (3, -4); at x = 0, y = 5.
11. B. Width 5 m, length 8 m. w(w + 3) = 40 gives (w + 8)(w - 5) = 0, so w = 5 and the length is 8 m. If your child chose D (width 8, length 5), they swapped the width and the length.
12. It lands when h = 0: -5t(t - 4) = 0, so t = 4 s. The vertex is halfway at t = 2, and h = -20 + 40 = 20, so the greatest height is 20 m.

Section 2: Shapes, measuring and space

1. D. Two sides and an angle not between them (SSA) SSA can fit two different triangles, so it does not prove congruence. If your child chose B (SAS), they mixed up the angle between two sides, which works, with an angle not between them.
2. The equal angle B is between the equal sides AB and BC, and E is between DE and EF. So it is SAS: two sides and the angle between them.
3. C. 125, 55, 55. The vertical angle is also 125 degrees, and the other two are 180 - 125 = 55 degrees. If your child chose A (55, 55, 55), they forgot that the opposite angle equals 125.
4. The two base angles are (180 - 40) / 2 = 70 degrees each. They are equal because, in an isosceles triangle, the angles opposite the equal sides are equal.
5. B. 15. The scale factor is 9 / 3 = 3, so x = 5 × 3 = 15. If your child chose A (11), they added 6 to 5 instead of multiplying by 3.
6. A. 15 m. The triangles are similar, so 3 / 2 = h / 10 and h = 10 × 3 / 2 = 15 m. If your child chose B (11 m), they added 10 + 1 instead of using the ratio.
7. The height 6 m is opposite the angle and the ramp 10 m is the hypotenuse, so sin A = 6 / 10 = 0.6. The angle whose sine is 0.6 is about 37 degrees.

8. The height is opposite the angle and 50 m is next to it, so use tan: height = 50 x tan 35 = 50 x 0.700 = 35. The tower is about 35 m tall.
9. The diagonal is the hypotenuse: 12^2 + 5^2 = 144 + 25 = 169, and the square root of 169 is 13. The walk is 13 m long.
10. Circumference = 2 x pi x 7 = 14 pi, about 44.0 m. Area = pi x 7^2 = 49 pi, about 153.9 square m. A quarter gives arc 14 pi / 4 = 7 pi / 2, about 11.0 m, and area 49 pi / 4, about 38.5 square m.
11. An inscribed angle is half the central angle on the same arc: 80 / 2 = 40 degrees. A diameter is a 180 degree arc, so its inscribed angle is 180 / 2 = 90 degrees.
12. C. 12 cm. A tangent is at right angles to the radius at T, so PT^2 = 13^2 - 5^2 = 144 and PT = 12 cm. If your child chose A (8 cm), they just subtracted 13 - 5 instead of using the right triangle.

Section 3: Data, graphs and chance

1. A. 1/12. The two events do not affect each other, so multiply: 1/2 × 1/6 = 1/12. If your child chose D (2/3), they added 1/2 + 1/6 instead of multiplying.
2. Independent means P(A) x P(B) = P(A and B). Here 0.4 × 0.5 = 0.2, which equals P(A and B), so yes, they are independent.
3. The first counter is put back, so both draws have 8 counters. P(red) = 5/8 each time, and 5/8 × 5/8 = 25/64.
4. C. 5/14. With no putting back, 7 counters are left and 4 are red: 5/8 × 4/7 = 20/56 = 5/14. If your child chose A (25/64), they forgot the first counter was not put back.
5. Swimming: 15 + 20 = 35 students. P(swimming) = 35 out of 50 = 35/50 = 7/10.
6. D. 4/5. There are 20 + 5 = 25 girls and 20 prefer swimming, so 20/25 = 4/5. If your child chose B (7/10), they used all 50 students instead of only the girls.
7. 35 students prefer swimming, and 20 of them are girls. So P(girl given swimming) = 20/35 = 4/7.
8. Compare P(swimming) = 35/50 = 0.7 with P(swimming given girl) = 20/25 = 0.8. They are different, so the two events are not independent.
9. B. 0.6. P(B given A) = P(A and B) / P(A) = 0.3 / 0.5 = 0.6. If your child chose C (0.3), they stopped at P(A and B) and did not divide by P(A).
10. D. 120. Order matters, so 6 choices for gold, 5 for silver and 4 for bronze: 6 × 5 × 4 = 120. If your child chose C (216), they let the same runner win more than one place.
11. 4 choices for the first place, 3 for the second, 2 for the third and 1 for the last: 4 × 3 × 2 × 1 = 24. Only one is OMAN, so the probability is 1/24.
12. Days are independent, so multiply: 0.9 × 0.9 = 0.81 for both on time. Late at least once is the opposite: 1 - 0.81 = 0.19.

Section 4: Numbers and calculating

1. A. 6√2. 36 is the biggest square that divides 72, so √72 = √36 x √2 = 6√2. If your child chose B (36√2), they forgot to take the square root of 36.
2. Simplify each: √50 = √25 x √2 = 5√2 and √18 = √9 x √2 = 3√2. Add the like roots: 5√2 + 3√2 = 8√2 cm.
3. Multiply under one root: √3 x √12 = √36. The square root of 36 is 6.

4. B. 2√3. Multiply top and bottom by √3: 6√3 / 3 = 2√3. If your child chose C (6√3), they forgot to divide by the new denominator 3.
5. A power of 1/2 means the square root, so 49^(1/2) = 7. A power of 1/3 means the cube root, so 8^(1/3) = 2.
6. C. 1/8. A negative power means flip, then 3/2 means the square root cubed: 1 / 4^(3/2) = 1 / 2^3 = 1/8. If your child chose D (8), they ignored the negative sign of the power.
7. The bottom number of the power is the root, so the cube root of 5 is 5^(1/3). In 7^(2/5), the 5 is the root and the 2 is the power: the fifth root of 7^2, which is the fifth root of 49.
8. D. 1/2 + √3. A rational number plus an irrational number is irrational, so 1/2 + √3 is irrational. If your child chose A (√2 x √8), they did not multiply the roots first, which gives √16 = 4.
9. A. 30 m/s. 1 km is 1000 m and 1 hour is 3600 s, so 108 × 1000 / 3600 = 30 m/s. If your child chose C (108 m/s), they forgot to change the units.
10. In litres: 3 × 10^4 × 10^3 = 3 × 10^7 litres a day. For 250 days: 3 × 10^7 × 2.5 × 10^2 = 7.5 × 10^9 litres.
11. The error is 250 - 245 = 5 litres. Percent error = 5 / 250 × 100 = 2%.
12. Change 2 km to 2000 m, then 2000 × 500 = 1,000,000 square m, which is 1 square km. Mixing km and m would multiply different units and give a wrong answer.